PUBLIC BUS LIVE TRACKING & CROWDING ESTIMATOR

Lab 1 – Requirements Engineering & UML Use-Case Modelling

Requirements Table + Use-Case Flow

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1. Requirements Table

The following table contains exactly five functional requirements and two non-functional requirements for the given public bus tracking scenario.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall ingest GPS coordinates from city buses every 5 seconds and update the estimated arrival time (ETA) for the upcoming stops.	High	During a test with a running bus, GPS data is received once every 5 seconds (with a tolerance of ±1 second). After the bus speed or position changes, the ETA for the next stop is updated within 10 seconds.	Live GPS tracking and ETA are the main purpose of the system.	Given in the problem statement. The timing has been made measurable for testing.
FR-002	Functional	The system shall allow a commuter to select a bus route or stop and see the current location and expected arrival time of the buses serving that stop.	High	For 20 test searches using valid routes or stops, the system displays the latest available bus location and ETA within 5 seconds for each bus with valid data.	Commuters need a simple way to know where the bus is and when it is likely to arrive.	The screen should also show when the displayed location is not recent.
FR-003	Functional	The system shall use passenger-count information from the bus ticketing sensors to calculate and display the current crowding level of each active bus.	High	For 100 test cases with known passenger counts, the crowding level shown by the system matches the expected level in all 100 cases.	Crowding information helps passengers decide which bus to take.	Crowding limits can be set according to the capacity of the bus.
FR-004	Functional	The system shall provide the fleet controller with a dashboard showing active buses, latest GPS update time, ETA status and crowding level.	High	When a bus has no GPS update for more than 15 seconds, or reaches the critical crowding limit, the dashboard shows an alert for that bus within 10 seconds.	The fleet controller needs one place to quickly check the condition of the buses.	The alert limits should be configurable.
FR-005	Functional	The system shall send an alert to a commuter when a selected bus is delayed beyond the configured delay limit or reaches the critical crowding level.	Medium	In 20 test cases where the delay or crowding limit is crossed, an alert is sent within 15 seconds. No alert should be sent when the limit is not crossed.	This helps commuters make a different travel choice when the selected bus is delayed or too crowded.	The commuter must have notifications enabled to receive the alert.
NFR-001	Nonfunctional	The tracking system shall support telemetry from up to 1,000 active buses at the same time without going beyond the required performance limits.	High	In a load test with 1,000 active bus streams sending data at the required rate, no more than 5% of telemetry messages are dropped and 95% of processing requests add no more than 2 seconds of latency.	The system should continue working when a large number of buses are active at the same time.	Given in the problem statement; should be checked using a simulated peak-load test.
NFR-002	Nonfunctional	The system shall protect bus location and passenger-count data and allow operational functions only to users with the required access.	High	All external API and telemetry connections use TLS, unauthenticated requests are rejected, and a Commuter account cannot access Fleet Controller functions in all authorization tests.	Location and operational data should not be available to people who are not authorized to see it.	Security testing should be completed before the system is released.

2. Actors

- **Commuter –** Uses the system to check bus location, ETA and crowding information.
- **Fleet Controller –** Monitors buses and responds to delays, stale data and crowding problems.
- **Bus GPS Unit –** Sends the current location and movement information of a bus.
- **Passenger Counting Sensor –** Sends passenger-count information used to estimate crowding.

3. Use Cases

- UC-01 – View Live Bus Status
- UC-02 – Check Stop ETA

- UC-03 – View Crowding Level
- UC-04 – Monitor Fleet
- UC-05 – Receive GPS Data
- UC-06 – Receive Passenger Count
- UC-07 – Send Service Alert
- UC-08 – Get Latest Bus Location
- UC-09 – Calculate Crowding

4. UML Relationships Used

- UC-01 View Live Bus Status «include» UC-08 Get Latest Bus Location.
- UC-03 View Crowding Level «include» UC-09 Calculate Crowding.
- UC-07 Send Service Alert «extend» UC-04 Monitor Fleet, because the alert is triggered only when a delay, stale-data or crowding condition occurs.

5. Use-Case Flow – UC-01 View Live Bus Status

Primary Actor	Commuter
Supporting Actors	Bus GPS Unit, Passenger Counting Sensor
Preconditions	The commuter can access the application. The selected route or stop is available. The tracking service is running.
Postconditions	The commuter sees the latest available bus location, ETA and crowding level. If data is not current, the system clearly says so.

Main Success Scenario

1. The commuter opens the bus tracking screen and selects a route or stop.
2. The system checks the selected route or stop and finds the buses currently serving it.
3. The system gets the latest GPS information for those buses.
4. The system calculates the expected arrival time for the selected stop.
5. The system gets the latest passenger count for each bus.
6. The system converts the passenger count into a crowding level using the configured limits.
7. The system shows the bus location, ETA and crowding level on the screen.
8. The system also shows the time at which the bus information was last updated.
9. The commuter checks the information and can select another route or stop if required.
10. The use case ends.

Alternate Flow – GPS Information Is Stale

1. At Step 3, the system finds that the latest GPS update for a bus is more than 15 seconds old.
2. The system marks that bus as “Data Stale” and shows the last update time instead of presenting the old position as live.
3. If none of the buses have fresh GPS information, the system displays “Live location unavailable”.
4. The commuter can try again after some time or select another route/stop.

Another Possible Exception – No Crowding Data

If the passenger-count sensor has not sent usable data, the system shows “Crowding data unavailable”. It still shows the bus location and ETA if those values are available.

6. Use-Case Flow – UC-01 View Live Bus Status

